Typed Judgment, Deterministic Enforcement: A Consent-Gated Gateway Architecture for Inter-Agent Collaboration over Messaging Channels

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The boundary we study

Messaging events are unstructured and may be duplicated, delayed, or replayed; agent protocols expect authenticated endpoints, structured requests, and task state.

- Free-text delegation lets model output become an invocation.
- Static rules cannot resolve natural-language ambiguity.

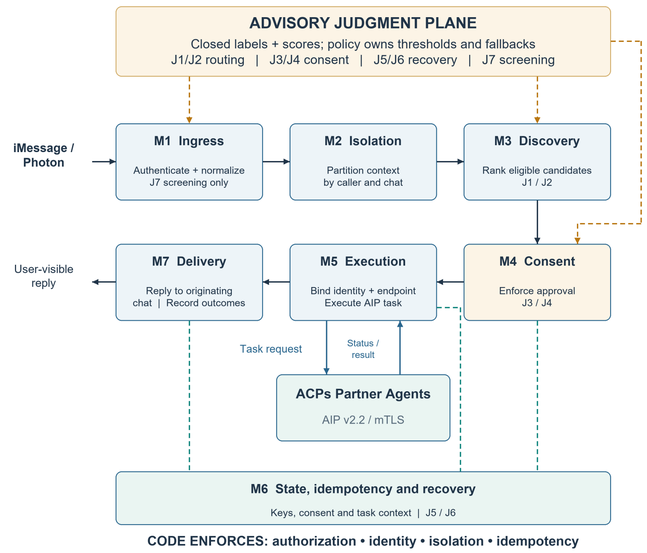
Design rule. The model answers seven closed questions (J1–J7); policy code owns thresholds and effects.

Seven modules, four invariants

M1 ingress authentication
M2 caller / conversation isolation
M3 discovery and routing
M4 consent and policy
M5 identity-bound execution
M6 state, idempotency, recovery
M7 delivery and telemetry

In code: authorization • identity • isolation • idempotency

Architecture



M1 ingress, M2 partition by caller and conversation, M3 discovery, M4 consent, M5 identity-bound execution, M6 state, M7 delivery.

Consent gate

Before consent: **0** / **40** partner actions.
Consents producing exactly one action: **36** / **36**.
Replayed consents: no extra action, identical stored reply.

Results: routing (130 requests, 3 deliveries)

Direct	48/48/50 of 50
Ambiguous	39/38/38 of 40
Out of scope	0 of 40

Generated target recommended; more than one agent when two capabilities fit; no partner when none advertises the request.

Results: invariants (paired ablations)

Arm	Control	Removed
Identity binding	0/30	30/30
Caller isolation	0/30	30/30
Idempotency	30 exec.	60 exec.

Same-code controls; each arm changes exactly one check.

Results: judgment layer

	Typed	Free text
Accuracy	94.1%	83.9%
Brier	0.095	0.266
ECE	0.026	0.144

280 items, 3 runs, human-verified labels.

Takeaway

Gateway cost